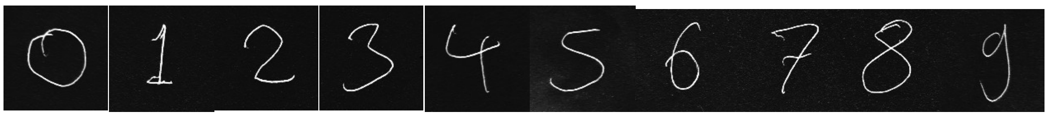
Deepa Sury

01.06.2025

Task 2.5

Part 1

Testing MNIST data-trained CNN on handwritten numbers



epochs = 50

batch_size = 16

n_hidden = 32

accuracy: 0.9917 - loss: 0.0302

True Value	**Predicted Value**
0	7
1	2
2	**2**
3	**3**
4	8
5	3
6	5
7	7
8	3
9	3

The model predicted 2 out of 10 numbers correctly.

Changing the model parameters.

- Changed the size of the second 2D convolution layer to 64
- Changed the dropout rate from 0.25 to 0.5

These changes are intended to improve the edge recognition of the model while preventing it from overfitting the training data.

Train data

accuracy: 0.9893 - loss: 0.0335

Test data evaluation

accuracy: 0.9930 - loss: 0.0288

True Value	**Predicted Value**
0	7
1	2
2	3
3	**3**
4	8
5	**5**
6	3
7	8
8	3
9	7

This model once again predicted only 2 numbers correctly.

Changed the size of the 2D convolution layers to 16 and 32.

True Value	**Predicted Value**
0	7
1	**1**
2	3
3	**3**
4	8
5	**5**
6	5
7	2
8	3
9	5

This time the model predicted 3 out of 10 numbers correctly.

Removed the second 2D convolution layer, as I suspected it was causing overfitting.

True Value	**Predicted Value**
0	7
1	8
2	**2**
3	**3**
4	**4**
5	**5**
6	8
7	2
8	**8**
9	3

This time the model correctly predicted 5 out of 10 numbers.

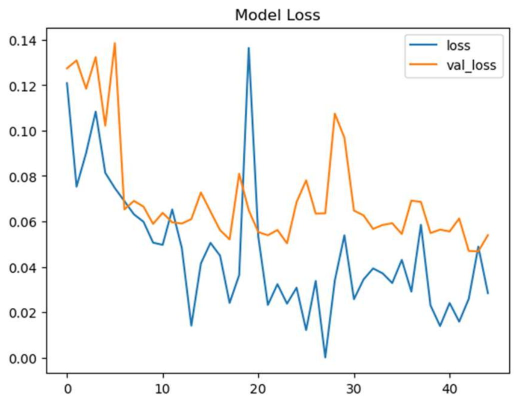
Part 2

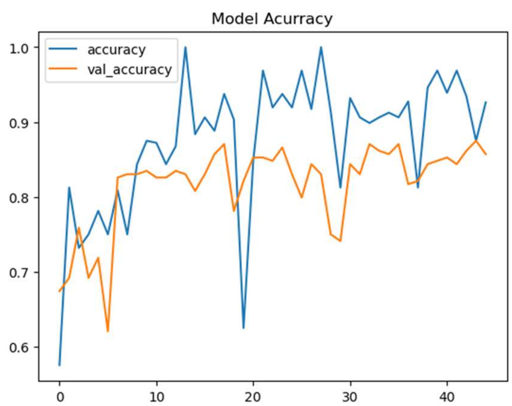
Training the Generator model

epochs =45

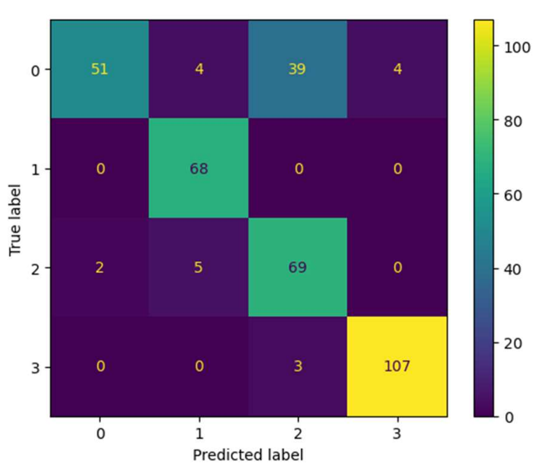
learning_rate=0.01

accuracy: 0.9315 - loss: 0.0266 - val_accuracy: 0.8571 - val_loss: 0.0539





Confusion matrix for weather image data



The classes correspond to the following weather conditions: Cloudy (0), Rain (1), Shine (2) and Sunrise (3)

Overall, the model showed a high rate of accuracy with these predictions. Sunrise (class 3) had the highest number of correct predictions, while cloudy (class 0) had the lowest.

The following images are 2 examples of Cloudy images predicted as other classes.

The first image, predicted as Rain, shows no rain, but contains the characteristic dark clouds associated with rainfall. The second image, predicted as Shine, shows a considerable amount of clear blue sky along with the clouds.

A GAN like this interprets images like these as arrays of individual colour codes (one number for each pixel comprising the image). The 2 conditions predicted most accurately also happen to contain distinctive colour palettes not present in other weather conditions – warm tones are mostly present during sunrise, bright blues are associated with sunshine. Rain and clouds, on the other hand, are predominantly identified by the frequent presence of grey, which makes it difficult to correctly classify images showing, for instance, a partly-cloudy sky, or an impending thunderstorm.

Incorrect Prediction - class: Cloudy - predicted: Rain[0.23311575 0.6953087 0.06878766 0.00278795]

Incorrect Prediction - class: Cloudy - predicted: Shine[0.36946824 0.09674808 0.5164023 0.01738137]

3 Ideas for Further Use of GANs in Weather Prediction

1. GANs can interpret weather satellite imagery over time to recognize patterns and predict upcoming precipitation or storms. One example of this can be found here, where a conditional GAN is trained on radar images of cloud cover to generate maps that predict rainfall.

2. Visualizing temperature, humidity and UV readings from measurement devices across a region, then using these visualizations to predict temperature changes.

   I found an example similar to this, using a GAN trained on historical temperature data to generate hourly temperature readings. More information on the TemperatureGAN can be found here.

3. For coastal regions, GANs can use image data from the water surface to identify wind conditions and assess the safety of marine travel in the area.

   Here is a similar example of weather predictions using a GAN trained on outdoor webcam images to generate easily-interpretable predictions of future weather conditions.